Aryan Tiwari

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Summary

Computational engineer and machine learning practitioner with specialized experience in time-series forecasting (GRU/RNNs) and medical image analysis (CNNs). Adept at building end-to-end data pipelines, handling noisy multivariate datasets, and deploying predictive models. Passionate about translating engineering methodologies—such as run-to-failure mechanical degradation modeling—into computational biology to study neurodegeneration and the aging brain.

Education

Vellore Institute of Technology
Bachelor of Technology in Computer Science and Engineering | **CGPA – 8.45** 2022 - 2026

Technical Skills

Machine Learning & Data Science: PyTorch, Scikit-learn, Pandas, NumPy, Matplotlib, OpenCV, Seaborn, Grad-CAM

Programming Languages: Python (4+ years), JavaScript (ES6+), SQL, Java, C++

Backend & Web Technologies: Node.js, Express.js, REST API Design

Databases & Data Stores: PostgreSQL, MySQL, MongoDB

Tools & Platforms: Git/GitHub, Docker, Google Colab, Jupyter Notebook, AWS, Render, Vercel

Internship

Infosys Springboard **Nov 2025 – Jan 2026**
Machine Learning Intern | Remote

- Designed and implemented an end-to-end supervised learning workflow for Remaining Useful Life (RUL) prediction using multivariate time-series sensor data (NASA C-MAPSS), effectively modeling progressive system degradation.
- Engineered a Gated Recurrent Unit (GRU) sequence model, implementing custom data preprocessing, feature scaling, and a sliding-window algorithm to capture long-term temporal dependencies.
- Conducted structured experimentation across datasets with varying operational complexities and fault modes, utilizing early stopping and model checkpointing to ensure robust generalization.

Edunet Foundation **Jan 2025 - Feb 2025**
Python Machine Learning Intern | Remote

- Developed a predictive modeling system using Random Forest to analyze complex environmental datasets (soil pH, NPK, climate), improving recommendation accuracy by 15%.
- Optimized the data preprocessing pipeline, reducing model training time by 30% through efficient feature engineering.
- Deployed the model via a Flask-based REST API and integrated the Google Gemini API to generate human-readable, natural-language explanations for the model's predictions, enhancing interpretability.

Projects

PulmoScan – Clinical Deep Learning for Pneumonia Detection Live Demo

- Engineered a deep learning diagnostic web application using a fine-tuned MobileNetV2 CNN, achieving 88.14% test accuracy for pneumonia classification on a chest X-ray dataset.
- Implemented **Grad-CAM (Gradient-weighted Class Activation Mapping)** using TensorFlow to provide Explainable AI (XAI), generating visual heatmaps to highlight the specific biological regions driving the model's predictions.
- Developed and deployed the end-to-end ML pipeline and Flask/Jinja web interface via RailwayCLI, efficiently handling medical image preprocessing, inference, and real-time visualization overlays.
- **Tech:** Python, TensorFlow 2.19, MobileNetV2, OpenCV (Grad-CAM), Flask, HTML/CSS, RailwayCLI

KalpChitran – AI Image Generation Platform Live Demo

- Engineered a full-stack AI image platform (MERN stack) integrating the FLUX.1-schnell model via the Hugging Face API to synthesize and render high-fidelity images from text prompts.
- Designed scalable Node.js/Express REST APIs to manage asynchronous image streaming, processing, and retrieval, demonstrating proficiency in handling heavy, unstructured visual data.
- Architected a robust backend deployed on Render, utilizing MongoDB Atlas for metadata persistence and Cloudinary for the secure, optimized cloud storage of image assets.
- **Tech:** React.js, Node.js, Express.js, MongoDB Atlas, Cloudinary, Hugging Face API (FLUX.1), Tailwind CSS.

Achievements

- Solved 900+ data structure and algorithm problems on LeetCode
- Ranked globally at 1132 in Weekly Contest 482 (Top 16%)

Certifications

- **IBM Data Science** Professional certificate by Coursera [IBM Data Science Certificate]
- **Oracle** Cloud Infrastructure 2025 Certified **Foundations Associate** [Oracle OCI Certificate]